Dr. D. Y. Patil Educational Federation's
Dr. D. Y. Patil College of Engineering & Innovation
Approved by AICTE, Recognized by Govt. of Maharashtra,
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
Accredited by NAAC with "A" Grade

Department of Computer Engineering

Program name :- Computer Engineering **Class:** S.Y. (B. Tech) **Div:-** B

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PRN No :-251110118 **Roll No:-** 12254 **Sem :-** 3rd **Course :-** BSFI (Online)

SMARTBRIDGE | SKILLWALLET

Personal Finance Advisor Bot

Project Description :-

The Personal Finance Advisor Bot is a personal finance management application designed to help individuals manage their money with clarity and confidence. It enables users to record monthly income, track daily expenses by category, plan budgets, monitor savings, and review their financial activity in one place.

The application is designed around a Flask-based web backend and a SQLite database to organize income, expenses, and budget records. Budget analysis can help users understand spending patterns and identify categories where they may be exceeding their planned limits. If configured in the working application, Gemini AI can provide natural-language explanations and budgeting suggestions.

The project aims to reduce the effort involved in manual financial tracking by presenting records and summaries in a structured interface. Possible future enhancements include spending trend analysis, goal-based savings tracking, and improved financial reports. The final feature set should reflect what has actually been implemented and tested.

Technology & AI :-

Frontend: HTML / CSS / JavaScript
Backend: Python / Flask

Database: SQLite
ORM: SQLAlchemy (if used in the implementation)
AI: Google Gemini API (if configured)
Development: Visual Studio Code
Security: Store API keys in environment variables; do not hard-code secrets

Key Features :-

- **Record monthly income and daily expenses**
- **Organize expenses into categories such as food, rent and transport**
- **Plan budgets and compare planned limits with actual expenses**
- **Review income, expenses and savings in monthly summaries**
- **Track savings and work towards financial goals**
- **Generate budget insights from recorded financial data**
- **Optional Gemini AI-assisted explanations and recommendations**
- **Simple web interface for personal finance management**
- **Potential future enhancement: spending trends and goal-based reports**

Scenarios :-

Scenario 1: A salaried professional records monthly income and expenses such as rent, food, transport and entertainment. The application summarizes spending by category and helps compare actual expenses with the planned budget, making it easier to review monthly savings.

Scenario 2: A college student enters a limited monthly allowance and records everyday purchases. By reviewing category totals and budget limits, the student can identify unnecessary spending and plan how much money to save.

Scenario 3: A freelancer records income received from different clients along with work-related expenses. Monthly summaries help the freelancer understand changes in income, plan for lower-income months and work towards an emergency fund.

Scenario 4: A household user records expenses such as groceries, utilities, education and healthcare. The monthly summary helps identify categories that exceed planned limits and supports more organized household budgeting.

Milestones :-

Milestone 1 – Project Planning & Requirement Analysis

Define the project's objective, identify target users, finalize the core finance-management features, and plan the application workflow.

Milestone 2 – Database Design

Plan the data structure for income, expenses, expense categories and budget records using SQLite; use SQLAlchemy only if it is part of the actual implementation.

Milestone 3 – Backend and Budget Analysis

Develop the Flask routes, validate user inputs, connect the database, and implement the calculations required for income, expenses, budgets and savings summaries.

Milestone 4 – Frontend and AI Assistance

Build the HTML, CSS and JavaScript interface. If Gemini is integrated, connect it securely to provide AI-assisted explanations or suggestions based on appropriate user-provided information.

Milestone 5 – Testing and Deployment

Test input validation, database operations, budget calculations and the user interface. Configure secrets safely and deploy the application only after the required features have been tested.

Conclusion :-

The Personal Finance Advisor Bot aims to make everyday money management more organized and understandable. By combining a web interface, Flask backend, SQLite storage and budget analysis, it can help users review spending habits and understand their monthly finances. AI-assisted recommendations are available only if the Gemini integration is configured. The project is intended for budgeting support and is not a substitute for professional financial advice.

Expected Outcome :-

The expected outcome is a web application that allows users to record income and expenses, organize spending by category, compare expenses with budget limits, and review monthly financial summaries. If implemented and tested, additional AI-assisted guidance can help explain spending patterns and suggest practical budgeting habits.